# Project Planning Phase

## Product Backlog, Sprint Planning, Stories & Story Points

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|---|---|
| **Date** | 7 March 2026 |
| **Team ID** | Abhay Nayakwdi, Prajwal Kotgonda,<br>Rohan Pawar, Sourav Kadam |
| **Project Name** | Heart Disease Analysis |
| **Maximum Marks** | 5 Marks |

## Product Backlog, Sprint Schedule & Estimation

The product backlog captures all user stories and tasks required to build the Heart Disease Analysis Tableau dashboard. Stories are organized into two sprints based on priority and dependency. Each story is assigned story points using the Fibonacci scale to estimate effort.

| Sprint | Functional Requirement (Epic) | Story No. | User Story / Task | Story Points | Priority | Team Member |
|---|---|---|---|---|---|---|
| **SPRINT 1 — Data Setup & Core Visualizations \| Total Story Points: 9** | | | | | | |
| Sprint1 | Data Collection & DB Setup | USN-1 | As a team, we can import Heart_new2.csv into MySQL, verify schema, and confirm all 18 columns are correctly mapped | 3 | High | Team Member |
| Sprint1 | Gender vs Heart Disease | USN-2 | As a user, I can view a stacked bar chart showing heart disease count by gender (Female: 2604, Male: 1896) | 2 | High | Team Member |
| Sprint1 | Age vs Heart Disease | USN-3 | As a user, I can view heart disease distribution across all 13 age categories segmented by gender | 2 | High | Team Member |
| Sprint1 | Smoking & Alcohol Impact | USN-4 | As a user, I can view how smoking and alcohol combinations affect heart disease count in a bar chart | 2 | High | Team Member |
| **SPRINT 2 — Advanced Visualizations, Dashboard & Publishing \| Total Story Points: 12** | | | | | | |

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|---|---|---|---|---|---|---|
| Sprint2 | BMI & Diabetic Analysis | USN-5 | As a user, I can view a treemap of age groups and average BMI values segmented by diabetic status | 2 | Medium | Team Member |
| Sprint2 | Race-wise Distribution | USN-6 | As a user, I can view a pie chart showing heart | 2 | Medium | Team Member |
| | | | disease prevalence across all 6 racial groups | | | |
| Sprint2 | Physical Activity & General Health | USN-7 | As a user, I can view physical activity donut chart (3,231 active vs 1,269 inactive) and general health bubble chart | 2 | Medium | Team Member |
| Sprint2 | Stroke & Disease Cooccurrence | USN-8 | As a user, I can view stroke vs diabetes scatter plot and stroke vs asthma/kidney/skin cancer bar chart | 2 | High | Team Member |
| Sprint2 | Dashboard & Story Assembly | USN-9 | As a user, I can view all 10 sheets in a combined Tableau Dashboard and Story titled Heartbeat of Data | 3 | High | Team Member |
| Sprint2 | Publish to Tableau Public | USN-10 | As a team, we can publish the final packaged workbook (.twbx) to Tableau Public for stakeholder access | 1 | High | Team Member |

## Project Tracker, Velocity & Burndown Chart

### Sprint Summary Tracker

| Sprint | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed | Sprint Release Date (Actual) |
|---|---|---|---|---|---|---|
| Sprint-1 | 14 | 7 Days | 4 March 2026 | 10 March 2026 | 14 | 10 March 2026 |
| Sprint-2 | 12 | 7 Days | 11 March 2026 | 17 March 2026 | TBD | TBD |

## Velocity Calculation

Velocity measures the amount of work a team completes during a sprint, expressed in story points per day. It is used to predict how much work can be completed in future sprints and helps in accurate sprint planning.

| Sprint-1 Velocity Metrics | Value |
|---|---|
| **Story Points Planned** | **14** |
| **Story Points Completed** | **14** |
| **Sprint Duration** | **7 Days** |
| **Velocity (Points/Day)** | **14 ÷ 7 = 2.0 story points/day** |
| **Sprint-2 Estimation (12 pts)** | **12 ÷ 2.0 = 6.0 days → Planned: 7 Days** |

Use of Velocity:

- To estimate future sprints more accurately based on past performance.
- Sprint-2 has 12 story points. Based on velocity of 2.0 points/day: 12 ÷ 2.0 = 6.0 days → planned as 7 days.
- Velocity helps identify team capacity and flag if a sprint is overloaded before it begins.

## Sprint-1 Burndown Chart Data

The burndown chart tracks the remaining work (story points) against elapsed time across Sprint-1. An ideal burndown line shows steady reduction from 14 points on Day 1 to 0 on Day 7. The table below shows the actual vs ideal burndown for Sprint-1.

| Day / Date | Ideal Remaining Points | Actual Remaining Points | Points Completed (Day) | Cumulative Completed | Status |
|---|---|---|---|---|---|
| Day 1 (4 Mar) | 14 | 12 | 2 | 2 | **On Track** |
| Day 2 (5 Mar) | 12 | 9 | 3 | 5 | **On Track** |
| Day 3 (6 Mar) | 9 | 7 | 2 | 7 | **On Track** |
| Day 4 (7 Mar) | 7 | 4 | 3 | 10 | **On Track** |

| | | | | | |
|---|---|---|---|---|---|
| Day 5 (8 Mar) | 4 | 2 | 2 | 12 | **On Track** |
| Day 6 (9 Mar) | 2 | 0 | 2 | 14 | **Completed** |
| Day 7 (10 Mar) | 0 | 0 | 0 | 14 | **Sprint Done** |

Burndown Chart Interpretation:

- Sprint-1 started with 14 story points on 4 March 2026 and was completed by 10 March 2026 with all 14 points delivered.
- The team maintained a steady pace of 2 story points per day throughout Sprint-1 with no blockers or delays.
- The actual burndown line tracked closely with the ideal burndown line, indicating good sprint planning and execution.
- Sprint-1 successfully delivered: MySQL setup, Gender vs Heart Disease chart, Age vs Heart Disease chart, and Smoking & Alcohol impact visualization.

## Story Points & Sprint Summary

| Sprint | Total Stories | Total Story Points | Duration | Status |
|---|---|---|---|---|
| Sprint-1 | 4 Stories (USN-1 to USN-4) | 14 Points | 7 Days | Completed |
| Sprint-2 | 6 Stories (USN-5 to USN-10) | 12 Points | 7 Days | In Progress |
| **TOTAL** | **10 Stories** | **26 Points** | **14 Days** | **—** |

## Definition of Done (DoD)

A User Story is considered DONE when all of the following criteria are met:

- Tableau visualization is created and matches the user story acceptance criteria.
- All filters (age, gender, race, lifestyle) work correctly on the visualization.
- Data values are verified against the MySQL database — no discrepancies.
- Chart labels, titles, and axis names are correctly formatted and readable.
- The visualization is added to the Tableau Dashboard and Story layout.
- The workbook is saved as a packaged .twbx file with data included.

- Dashboard is published to Tableau Public and accessible via public URL.